

Assignment 2: Web Development Team Management

Task 1: Basic Setup for ProjectX

1. Create and navigate:

```
mkdir /var/www/ProjectX  
cd /var/www/ProjectX
```

2. Create files:

```
touch index.html app.py README.md
```

3. Verify directory:

```
pwd
```

4. List file details:

```
ls -l
```

5. Disk usage:

```
df -h
```

6. File content:

```
echo "Welcome to ProjectX" > README.md  
cat README.md
```

Output:

```
ubuntu@ip-172-31-93-65:~$ sudo mkdir /var/www/ProjectX  
ubuntu@ip-172-31-93-65:~$ cd /var/www/ProjectX  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ touch index.html app.py README.md  
touch: cannot touch 'index.html': Permission denied  
touch: cannot touch 'app.py': Permission denied  
touch: cannot touch 'README.md': Permission denied  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ sudo !!  
sudo touch index.html app.py README.md  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ pwd  
/var/www/ProjectX  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ ls -l  
total 0  
-rw-r--r-- 1 root root 0 Apr 27 20:32 README.md  
-rw-r--r-- 1 root root 0 Apr 27 20:32 app.py  
-rw-r--r-- 1 root root 0 Apr 27 20:32 index.html  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root        6.8G  2.3G  4.5G  35% /  
tmpfs            479M   0  479M   0% /dev/shm  
tmpfs            192M 904K  191M   1% /run  
tmpfs             5.0M   0   5.0M   0% /run/lock  
/dev/xvda16      881M 137M  683M  17% /boot  
/dev/xvda15      105M  6.1M   99M   6% /boot/efi  
tmpfs             96M  12K   96M   1% /run/user/1000  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ echo "Welcome to ProjectX" | sudo tee README.md  
Welcome to ProjectX  
ubuntu@ip-172-31-93-65:/var/www/ProjectX$ cat README.md  
Welcome to ProjectX
```

Task 2: Permissions for Developers

1. Create group and users:

```
groupadd devteam
useradd bhatti
useradd malik
usermod -aG devteam bhatti
usermod -aG devteam malik
```

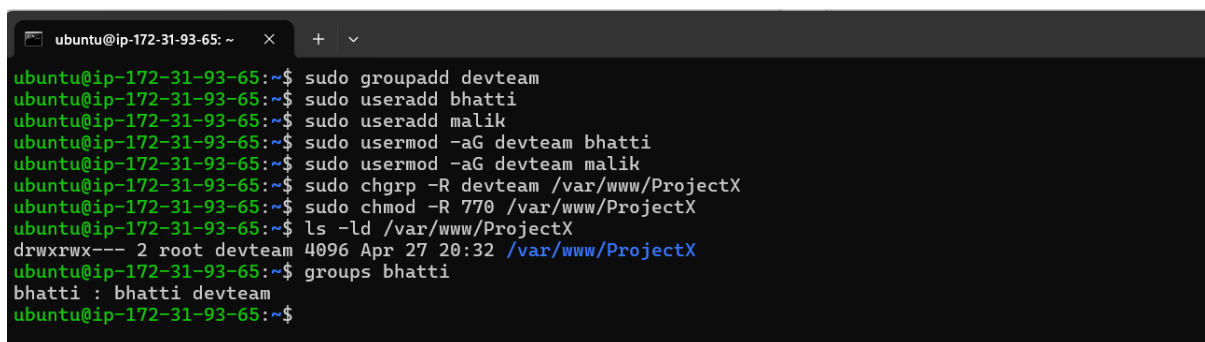
2. Change group ownership:

```
chgrp -R devteam /var/www/ProjectX
chmod -R 770 /var/www/ProjectX
ls -ld /var/www/ProjectX
```

3. Check groups:

```
groups bhatti
```

Output:



```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ sudo groupadd devteam  
ubuntu@ip-172-31-93-65:~$ sudo useradd bhatti  
ubuntu@ip-172-31-93-65:~$ sudo useradd malik  
ubuntu@ip-172-31-93-65:~$ sudo usermod -aG devteam bhatti  
ubuntu@ip-172-31-93-65:~$ sudo usermod -aG devteam malik  
ubuntu@ip-172-31-93-65:~$ sudo chgrp -R devteam /var/www/ProjectX  
ubuntu@ip-172-31-93-65:~$ sudo chmod -R 770 /var/www/ProjectX  
ubuntu@ip-172-31-93-65:~$ ls -ld /var/www/ProjectX  
drwxrwx--- 2 root devteam 4096 Apr 27 20:32 /var/www/ProjectX  
ubuntu@ip-172-31-93-65:~$ groups bhatti  
bhatti : bhatti devteam  
ubuntu@ip-172-31-93-65:~$
```

Task 3: Ownership

1. Change ownership:

```
chown -R bhatti:devteam /var/www/ProjectX
```

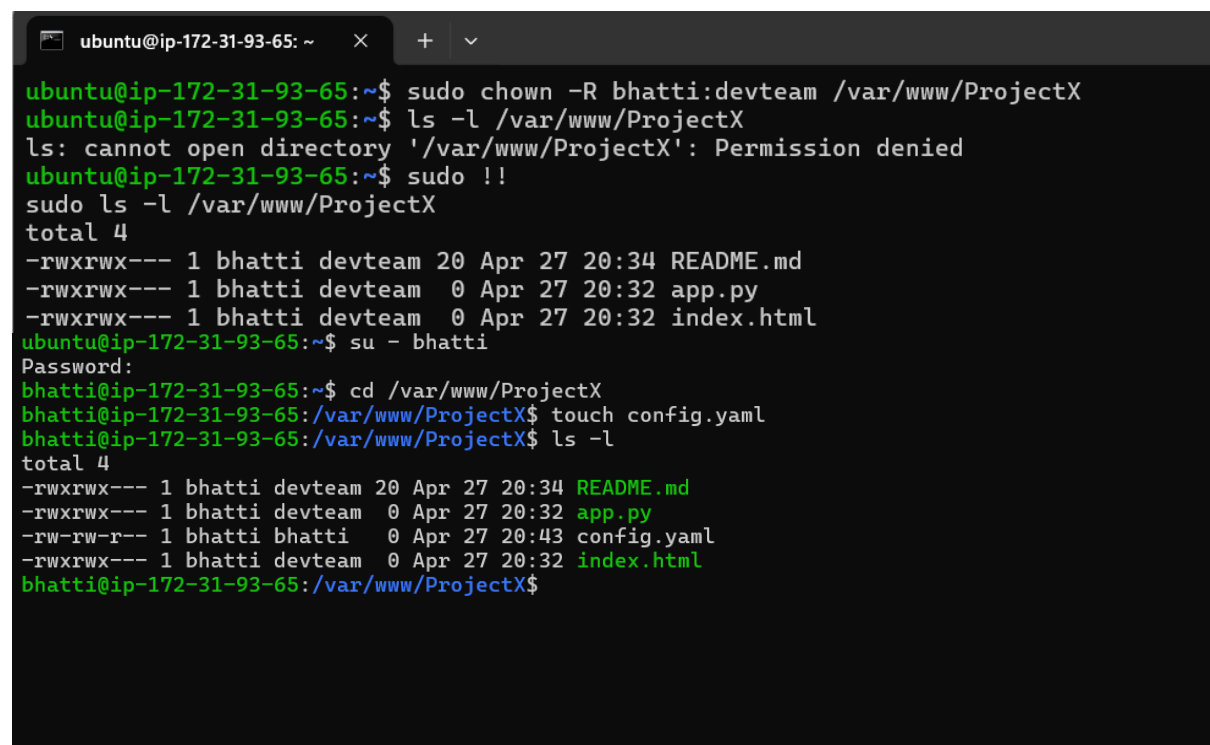
2. Verify:

```
ls -l /var/www/ProjectX
```

3. Login and add config:

```
su - bhatti
cd /var/www/ProjectX
touch config.yaml
ls -l
```

Output:



```
ubuntu@ip-172-31-93-65: ~ × + v
ubuntu@ip-172-31-93-65:~$ sudo chown -R bhatti:devteam /var/www/ProjectX
ubuntu@ip-172-31-93-65:~$ ls -l /var/www/ProjectX
ls: cannot open directory '/var/www/ProjectX': Permission denied
ubuntu@ip-172-31-93-65:~$ sudo !!
sudo ls -l /var/www/ProjectX
total 4
-rwxrwx--- 1 bhatti devteam 20 Apr 27 20:34 README.md
-rwxrwx--- 1 bhatti devteam  0 Apr 27 20:32 app.py
-rwxrwx--- 1 bhatti devteam  0 Apr 27 20:32 index.html
ubuntu@ip-172-31-93-65:~$ su - bhatti
Password:
bhatti@ip-172-31-93-65:~$ cd /var/www/ProjectX
bhatti@ip-172-31-93-65:/var/www/ProjectX$ touch config.yaml
bhatti@ip-172-31-93-65:/var/www/ProjectX$ ls -l
total 4
-rwxrwx--- 1 bhatti devteam 20 Apr 27 20:34 README.md
-rwxrwx--- 1 bhatti devteam  0 Apr 27 20:32 app.py
-rw-rw-r-- 1 bhatti bhatti   0 Apr 27 20:43 config.yaml
-rwxrwx--- 1 bhatti devteam  0 Apr 27 20:32 index.html
bhatti@ip-172-31-93-65:/var/www/ProjectX$
```

Task 4: System Monitoring

1. Resource usage:

top

Output:

```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ top  
top - 20:44:42 up 1 day, 14:05, 1 user, load average: 0.00, 0.00, 0.00  
Tasks: 101 total, 1 running, 100 sleeping, 0 stopped, 0 zombie  
%Cpu(s):  0.3 us,  0.0 sy,  0.0 ni, 89.8 id,  0.0 wa,  0.0 hi,  0.0 si,  9.9 st  
MiB Mem :  957.4 total,  187.7 free,  375.3 used,  588.9 buff/cache  
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  582.1 avail Mem  


| PID | USER | PR  | NI  | VIRT  | RES   | SHR  | S | %CPU | %MEM | TIME+   | COMMAND                     |
|-----|------|-----|-----|-------|-------|------|---|------|------|---------|-----------------------------|
| 1   | root | 20  | 0   | 22508 | 13952 | 9728 | S | 0.0  | 1.4  | 0:07.08 | systemd                     |
| 2   | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.01 | kthreadd                    |
| 3   | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | pool_workqueue_release      |
| 4   | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-rcu_g             |
| 5   | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-rcu_p             |
| 6   | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-slub_             |
| 7   | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-netns             |
| 10  | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/0:0H-events_highpri |
| 12  | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-mm_pe             |
| 13  | root | 20  | 0   | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | rcu_tasks_rude_kthread      |
| 14  | root | 20  | 0   | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | rcu_tasks_trace_kthread     |
| 15  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.49 | ksoftirqd/0                 |
| 16  | root | 20  | 0   | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.91 | rcu_sched                   |
| 17  | root | rt  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.83 | migration/0                 |
| 18  | root | -51 | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | idle_inject/0               |
| 19  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | cpuhp/0                     |
| 20  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | kdevtmpfs                   |
| 21  | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-inet_             |
| 23  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | kauditd                     |
| 24  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.03 | khungtaskd                  |
| 25  | root | 20  | 0   | 0     | 0     | 0    | S | 0.0  | 0.0  | 0:00.00 | oom_reaper                  |
| 27  | root | 0   | -20 | 0     | 0     | 0    | I | 0.0  | 0.0  | 0:00.00 | kworker/R-write             |


```

2. Running processes:

ps aux | grep ProjectX

Output:

```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ ps aux | grep ProjectX  
ubuntu    26382  0.0  0.2  7080  2048 pts/0    S+   20:45   0:00 grep --color=auto ProjectX
```

3. View system logs:

tail -n 50 /var/log/syslog

Output:

```
ubuntu@ip-172-31-93-65:~$ tail -n 50 /var/log/syslog
2025-04-27T20:30:18.930988+00:00 ip-172-31-93-65 PackageKit: daemon start
2025-04-27T20:30:18.936119+00:00 ip-172-31-93-65 systemd[1]: Starting polkit.service - Authorization Manager...
2025-04-27T20:30:18.960255+00:00 ip-172-31-93-65 udisksd[26058]: udisksd daemon version 2.10.1 starting
2025-04-27T20:30:18.996775+00:00 ip-172-31-93-65 dbus-daemon[578]: [system] Activating via systemd: service name='org.freedesktop.PolicyKit1' unit='polkit.service' requested by ':1.189' (uid=0 pid=26058 comm="/usr/libexec/udisksd" label="unconfined")
2025-04-27T20:30:19.003554+00:00 ip-172-31-93-65 multipathd[26064]: multipathd v0.9.4: start up
2025-04-27T20:30:19.016339+00:00 ip-172-31-93-65 multipathd[26064]: reconfigure: setting up paths and maps
2025-04-27T20:30:19.016401+00:00 ip-172-31-93-65 systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
2025-04-27T20:30:19.021261+00:00 ip-172-31-93-65 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
2025-04-27T20:30:19.050036+00:00 ip-172-31-93-65 polkitd[26062]: Started polkitd version 124
2025-04-27T20:30:19.055069+00:00 ip-172-31-93-65 dbus-daemon[578]: [system] Successfully activated service 'org.freedesktop.PolicyKit1'
2025-04-27T20:30:19.057217+00:00 ip-172-31-93-65 systemd[1]: Started polkit.service - Authorization Manager.
2025-04-27T20:30:19.064450+00:00 ip-172-31-93-65 systemd[1]: Starting ModemManager.service - Modem Manager...
2025-04-27T20:30:19.076908+00:00 ip-172-31-93-65 systemd[1]: Started packagekit.service - PackageKit Daemon.
2025-04-27T20:30:19.100634+00:00 ip-172-31-93-65 systemd[1]: Started udisksd.service - Disk Manager.
2025-04-27T20:30:19.105460+00:00 ip-172-31-93-65 ModemManager[26092]: <msg> ModemManager (version 1.23.4) starting in system bus...
2025-04-27T20:30:19.110077+00:00 ip-172-31-93-65 udisksd[26058]: Acquired the name org.freedesktop.UDisks2 on the system message bus
2025-04-27T20:30:19.141369+00:00 ip-172-31-93-65 systemd[1]: Started ModemManager.service - Modem Manager.
2025-04-27T20:30:26.856556+00:00 ip-172-31-93-65 systemd[1]: systemd-timedated.service: Deactivated successfully.
2025-04-27T20:30:44.974170+00:00 ip-172-31-93-65 systemd[1]: fwupd.service: Deactivated successfully.
2025-04-27T20:35:01.151735+00:00 ip-172-31-93-65 CRON[26177]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1 1)
```

```
d:
2025-04-27T20:36:06.753802+00:00 ip-172-31-93-65 amazon-ssm-agent.amazon-ssm-agent[952]: 2025-04-27 20:36:06.6910 ERROR EC2RoleProvider Failed to connect to Systems Manager with SSM role credentials. error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 864981713828
2025-04-27T20:36:06.753945+00:00 ip-172-31-93-65 amazon-ssm-agent.amazon-ssm-agent[952]: #011status code: 400, request id: 1b6d6788-5f86-43b1-b139-8805bdc2f99f
2025-04-27T20:39:10.675984+00:00 ip-172-31-93-65 adduser[26267]: Adding user 'bhatti' ...
2025-04-27T20:39:10.678492+00:00 ip-172-31-93-65 adduser[26267]: Selecting UID/GID from range 1000 to 59999 ...
2025-04-27T20:39:10.681180+00:00 ip-172-31-93-65 adduser[26267]: Adding new group 'bhatti' (1005) ...
2025-04-27T20:39:10.711415+00:00 ip-172-31-93-65 adduser[26267]: Adding new user 'bhatti' (1005) with group 'bhatti' (1005) ...
2025-04-27T20:39:10.774504+00:00 ip-172-31-93-65 adduser[26267]: Creating home directory '/home/bhatti' ...
2025-04-27T20:39:10.777939+00:00 ip-172-31-93-65 adduser[26267]: Copying files from '/etc/skel' ...
2025-04-27T20:39:33.835220+00:00 ip-172-31-93-65 adduser[26267]: Adding new user 'bhatti' to supplemental / extra groups 'users' ...
2025-04-27T20:39:33.837211+00:00 ip-172-31-93-65 adduser[26267]: Adding user 'bhatti' to group 'users' ...
2025-04-27T20:40:09.973914+00:00 ip-172-31-93-65 systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2025-04-27T20:40:09.985243+00:00 ip-172-31-93-65 systemd[1]: sysstat-collect.service: Deactivated successfully.
2025-04-27T20:40:09.985391+00:00 ip-172-31-93-65 systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
2025-04-27T20:45:01.158960+00:00 ip-172-31-93-65 CRON[26379]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1 1)
ubuntu@ip-172-31-93-65:~$
```